MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, GALLIUM ARSENIDE, MIXER TYPE 1N5764, 1N5764M and 1N5764MR

1. SCOPE

1.1 Scope. This specification covers the detail requirements for Gallium Arsenide, Low Barrier, High Sensitivity, Schottky-Barrier Semiconductor Diode, for use as a first detector in Ku-band equipment.

1.2 Ratings and characteristics.

	$\overline{Fo}$	1/G	L	N	SWR	Burnout R.F Power
	db	ohms	db	ratio	ratio	Watts
Minimum	---	400	---	---	---	1.5
Maximum	7.0	600	5.0	1.6	1.5	

OPERATING AMBIENT TEMPERATURE: -65° to +150°C

STORAGE TEMPERATURE: -65° to +150°C

BAROMETRIC PRESSURE, REDUCED: 8 min Hg

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

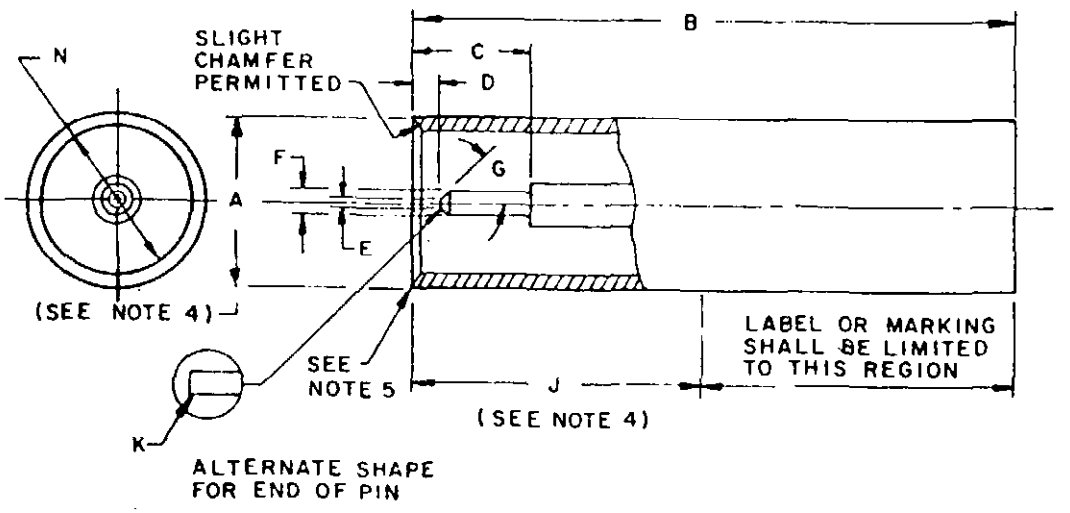
STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

DRAWINGS

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:11Z
Check the source to verify that this is the current version before use.



DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.215	.220	5.46	5.58
B	.734	.766	18.64	19.45
C	.147		3.73	
D	.011	.028	.28	.71
E	.007	.017	.18	.43
F	.031	.033	.78	.84
G	42°	48°	42°	48°
J	.406		10.31	
K	.007	.017	.18	.43
N	.179	.189	4.55	4.80

NOTES:
1. Finish: .0002"(.005 mm) tin plate over nickel flash, or .001"(.025 mm) gold plate or .0001"(.0025 mm) silver plate.
2. Axis of center conductor(pin) not to deviate from axis of outer conductor referred to its outside diameter more than .004"(.101 mm).
3. Forward polarity units shall have the cathode connected to the center pin. Reversed polarity units shall have the anode connected to the center pin.
4. Outside diameter, .215"(5.46 mm) to .220"(5.58 mm), applies for length of dimension J.
5. This device shall be free of sharp edges and burrs.

FIGURE 1. Semiconductor device, diode, type 1N5769, suffix R (reverse polarity), suffix M (matched forward pair), and suffix MR (matched forward and reverse).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:11Z
Check the source to verify that this is the current version before use.

Defense Electronics Supply Center

D65084	- Test Holder for Microwave Diode Type 1N78.
B66054	- Adapter for Burnout Test.
C66058	- Burnout Tester for Microwave Diodes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements for semiconductor diodes shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500 and as follows:

1/G - - - - - Reciprocal of conductance (IF impedance).
SWR - - - - - Standing wave ratio.
P - - - - - - Power from local oscillator.
P_K - - - - - Peak RF power.
P_{rr} - - - - - Pulse repetition rate.
f_o - - - - - Frequency of operation.

3.3 Design, construction, and dimensions. The semiconductor diode shall be of design, construction and physical dimensions specified in figure 1. Forward polarity units shall have the cathode connected to the center pin. Reversed polarity units shall have the anode connected to the center pin.

3.3.1 Plating. The diode shall be plated as specified in figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The marking shall be as specified in MIL-S-19500.

3.5.1 Matched diodes. The "M" suffix for matched diodes shall be omitted in the type designation on each device. Matched diodes meeting the requirements of this specification shall be packaged with a statement to the effect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:11Z
Check the source to verify that this is the current version before use.

4.2 Test Conditions. Unless otherwise specified herein the Test Conditions, when applicable, shall be as follows:

P = 1.0 mW ± 5%
f_o = 16.0 GHZ ± 10 MC
Zm = 500 ± 1% ohms ± J_o ohms
R_1 = 100 ± 1% ohms

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in Tables I, II, and III, except the matched-pair requirements in subgroup 3, Table I.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Groups A, B, and C.

4.4.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in Table I.

4.4.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in Table II.

4.4.3 Group C inspection. Group C inspection shall consist of the examinatic and tests specified in Table III.

4.5 Method of examination and test. Methods of examination and tests shall be as specified in Tables I, II , and III, and as follows:

4.5.1 High-temperature operation. The semiconductor diode shall be placed in the mixer holder. The ambient temperature of the diode, with test conditions specified for the overrall noise figure shall be raised to + 150°C and maintained at this temperature until equilibrium is reached. The Fo shall then be determined and shall be less then 10. dB. The temperature shall then be returned to 25±3°C at which time Fo shall be less than the specified limit.

4.5.2 Matched pair, "M" suffix (forward polarity). The matched forward pair (M suiffix) shall consist of two diodes, tested to requirements of subgroup 4, Table I, having the cathode connected to the center pin.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:11Z
Check the source to verify that this is the current version before use.

4.5.3 Matched pair, "MR" suiffix (forward and reverse polarity). The matched forward and reverse pair (MR suiffix) shall consist of two diodes, tested to requirements of subgroup 4 Table I: one diode having the cathode connected to the center pin (forward polarity) and the second diode having the anode connected to the center pin (reverse polarity).

4.5.4 Matched pair, Semiconductor devices shall be considered matched pairs when the following parameters; L, NR_o, and Zif are within specified limits, over the entire 0.5 to 2.0 MW local oscillator range.

4.5.5 Overall Noise Figure. The following parameters LC, NR_o and NF_o may be determined by the measurement of any suitable combination and the actual measured NF_{IF}. When NF_o is to be determined by calculation, where NF_{IF} is assumed at 1.5 ± 25 db, the measurement of both LC and NR_o parameter shall be required.

4.5.5.1 Mixer Holder. The mixer holder, per drawing D-65084 or JAN Drawing 201, when terminated with the matched load, per drawing C-65101 whall have a maximum vswr of 1.05:1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:11Z
Check the source to verify that this is the current version before use.

Table I. Group A Inspection.

Method MIL-STD-750	Examination or test	Details	LTPD	Symbol	Limits Min	Limits Max	Units
	Subgroup 1		3				
2071	Visual and Mechanical Examination						
	Subgroup 2		3				
	R.F. Burnout	P=1.5 Watts f=16 GHZ tp=3 nanosec. Prr=1000 min. t=1 min					
4136	Standing Wave Ratio			SWR		1.5	Ratio
4126	Overall (average) Noise (See 4.5.5 & 4.5.5.1) Figure			$\overline{Fo}$		7.0	db
4126	Overall (average) Noise (See 4.5.5 & 4.5.5.1) Figure	Except: P=0.2 mW		$\overline{Fo}$		9.0	db
	Subgroup 3		3				
4116	I.F. Conductance			1/G	400	600	ohms
4151	Rectified Current			I_o		0.7	mAdc
	Subgroup 4		3				
	Matched Pair Requirements: (See 4.5.2, 4.5.3 and 4.5.4)						
4101	Conversion Loss Unbalance			ΔL		0.3	db
4121	Output Noise Ratio Unbalance			ΔN		0.2	
4116	I.F. Conductance Unbalance			(1/G)		20	ohms

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:11Z
Check the source to verify that this is the current version before use.

Table II. Group B Inspection.

Method MIL-STD-750	Examination or test	Details	LTPD	Symbol	Limits Min	Limits Max	Units
	Subgroup 1						
2066	Physical dimensions	(See Figue 1) Dim. A,C,D F, and N. (Dim. E,G,K, and note 1 are for qualification only.)	7 20	---	---	---	---
	Subgroup 2		7				
1051	Thermal shock (temperature cycling)	Test Condition F		---	---	---	---
1021	Moisture resistance	Omit initial conditioning		---	---	---	---
	End points:						
4126	Overall(average noise Figure (see 4.2, 4.5.5 & 4.5.5.1)			$\overline{Fo}$	---	8.0	dB
	Subgroup 3		7				
2016	Shock	Nonoperating;500 G; t=1.0 msec;5 blows in each orientation: X_1, Y_1, and Y_2		---	---	---	---
2056	Vibration, variable frequency	Nonoperating: 15 G, 50 to 2,000 Hz		---	---	---	---
2006	Constant acceleration	Nonoperating; 20,000 G, X_1, Y_1, and Y_2 orientations		---	---	---	---
	End points: (Same as subgroup 2)						
	Subgroup 4		7				
---	High temperature operation: (see 4.5.1)	T_A=150°C		---	---	10.0	dB
4126	Overall (average noise figure (see 4.2, 4.5.5 & 4.5.5.1)			Fo	---	8.0	dB

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:11Z
Check the source to verify that this is the current version before use.

Table II. Group B Inspection - Continued

MIL-STD-750	Examination or test	Details	LTPD	Symbol	Limits Min	Limits Max	Units
	Subgroup 5		λ=10				
1031	High-temperature life (non-operating)	T_A=150°C		---	---	---	---
	End points: (Same as Subgroup 2)						

Table III. Group C Inspection

MIL-STD-750	Examination or test	Details	LTPD	Symbol	Limits Min	Limits Max	Units
	Subgroup 1		10				
4101	Conversion loss (See 4.2)			L	---	5.0	dB
4121	Output noise ratio (See 4.2)	R_e=300 ohms		N	---	1.6	ratio

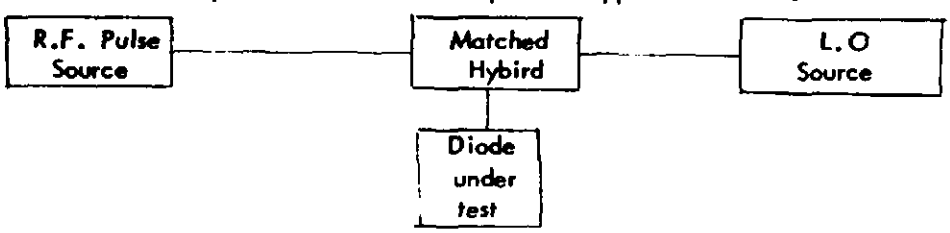
5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 The notes specified in MIL-S-19500 are applicable to this specification.

6.2 RF Pulse Test. The RF Pulse Test will be done in D-65084 or JAN Drawing number 201 test Holder, with RL = 100 ohms per the appended circuit.



6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity
Army-EL

(Project 5961-A386

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:11Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:11Z
Check the source to verify that this is the current version before use.